LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to sonobuoy system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to sonobuoy system. Purpose and technical details of active and passive sonobuoy.
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of sonobuoy types and signal transmission principles; Safety: No live equipment handling; Supervision: Instructor monitors understanding.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of a sonobuoy?
(b) Define the basic theory of communication.
(c) List two types of sonobuoys based on operation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of communication • Sonobuoy system overview • Purpose of sonobuoys • Signal transmission basics • Receiver and transmitter roles	L & Ds	PPT, IB	15 min
(b)	• Active sonobuoy technical details • Passive sonobuoy technical details • Comparison of active and passive • Frequency ranges used • Detection range factors	L & Ds	PPT, PDF	20 min
(c)	• Sonobuoy deployment procedures • Signal processing basics • Data interpretation concepts • System limitations • Maintenance overview	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between active and passive sonobuoys.
(b) Describe how a sonobuoy transmits data to the aircraft.

12. Home Assignment and Follow Up	Prepare a brief note on the communication theory concepts discussed in this lesson to be used in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to communication system. Purpose, technical details and types of various communication systems.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min